

12 (**short**)

- (a) Use Ampère's law to derive an expression for the magnetic field intensity B at a distance d from an infinite straight wire carrying a current I . [5]
- (b) As shown in Fig. 9, three orthogonal wires are each carrying a current I and cross without touching. Using the coordinate system in Fig. 9, and assuming that the currents flow in the positive directions, what is the magnetic field strength and direction at each of the four points (d, d, d) , $(-d, -d, -d)$, $(d, d, 0)$ and $(-d, -d, 0)$, where $d > 0$? [5]

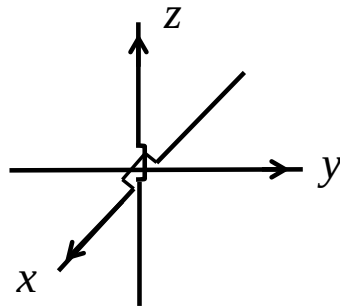


Fig. 9

END OF PAPER

Answers

Section A

Q1. (b) $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$; (c) gain = -8.33 , $R_{\text{out}} = 1.67 \text{ k}\Omega$; (d) $C_0 = 2.39 \times 10^{-6} \text{ F}$

Q2. (a) $R_2 = 3 \text{ M}\Omega$; (b) $R_{\text{in}} = 10 \text{ k}\Omega$; (c) $C_1 = 5.3 \text{ pF}$; (d) gain = 296

Q3. $Z_{\text{tot}} = 4 \Omega$, $I = 60 \text{ A}$

Q5. $I = 1 \text{ A}$

Section B

Q6. (b) $Z_1 = Y_1$, $Z_2 = Y_2 \bar{Y}_1 + \bar{Y}_2 Y_1$, $Z_3 = Y_2 \bar{Y}_3 + Y_1 \bar{Y}_3 + Y_3 \bar{Y}_2 \bar{Y}_1$, $Z_4 = Y_2 \bar{Y}_3 + Y_1 \bar{Y}_3$

Q7. (a) $Q_1 = S + \bar{Q}_2$, $Q_2 = R + \bar{Q}_1$

(b) master/slave configuration with clock, preset/clear, inputs both 1 toggles output

(c) depends on the delays present

Q8. (b) 509 cycles, 341 cycles

Q9. (a) $P_4 = \overline{\overline{X_1 + X_2}}$, $P_1 = \overline{\overline{\overline{X_1 + Y_1} + \overline{X_1 + Y_1}}}$

(b) $J_A = X_1 Q_B$, $J_B = \bar{X}_1$

Section C

Q10. (b) $V > d[2\rho g/\epsilon_0]^{1/2}$

Q11. (b) $H = 1000 \text{ A.m}^{-1}$, 600 A.turns; (c) 2188 A.turns; (d) 398 N

Q12. (b)

(d, d, d) $\mathbf{B} = B_d(0, 0, 0)$

Magnitude: 0

$(-d, -d, -d)$ $\mathbf{B} = B_d(0, 0, 0)$

Magnitude: 0

$(d, d, 0)$ $\mathbf{B} = B_d(-1/2, 1/2, 0)$

Magnitude: $B_d/\sqrt{2}$

$(-d, -d, 0)$ $\mathbf{B} = B_d(1/2, -1/2, 0)$

Magnitude: $B_d/\sqrt{2}$